

3(a)	$F = \rho g V$ $V = 4 / 3 \times \pi \times (2.1 \times 10^{-3})^3 = 3.88 \times 10^{-8} \text{ m}^3$ $\rho = 4.8 \times 10^{-4} / 9.81 \times V = 1300 \text{ kg m}^{-3}$	 [1] [1]
(b)(i)	W downwards > U upwards > F_v upwards. total length up = total length down	[1]
(ii)	$F_v = 7.2 \times 10^{-4} - 4.8 \times 10^{-4} = 2.4 \times 10^{-4} \text{ (N)}$ velocity = $2.4 \times 10^{-4} / (17 \times 2.1 \times 10^{-3}) = 6.7 \times 10^{-3} \text{ m s}^{-1}$	 [1] [1]

4 (a)	From infinity to Mars $ KE \text{ gain} = GPE \text{ loss} $	
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